# MANAS SHUKLA

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## EXPERIENCE

*Senior Systems Engineer,* **Sensata Technologies** | Attleboro, MA, USA **Jan 2022 – Mar 2025**

- Led software development for multimillion-dollar, multi-station complex machinery performing operations such as calibration, characterization, pick and place, etc. by designing control logic and operational sequences in compliance with the URS.
- Developed control systems using various PLC platforms, enhancing automation processes and improving equipment cycle time by roughly 15% while reducing operational errors by ~20% through a modular architecture design philosophy.
- Designed HMI interfaces using WinForms and TwinCAT; built SCADA dashboards for intuitive equipment control & monitoring.
- Created and optimized SQL database queries, including table creation and data management, leveraged Python scripting with SQL to interface equipment with the MES, enhancing data-driven decision-making, equipment traceability, and performance tracking.
- Configured servo drives, actuators, robots, power supplies, VFDs, etc., using protocols like Modbus, EtherCAT, and Ethernet/IP.
- Implemented safety systems for gantries, pneumatics, robots, and conveyors, ensuring compliance with industry standards.
- Managed project schedules, costs, scope, and objectives while coordinating with suppliers and cross-functional team members across the globe to ensure seamless automation integration and scalability from conceptual development through installation.
- Conducted statistical analyses (GRR and ANOVA) to validate manufacturing process improvements, ensuring equipment accuracy.

*Machine Learning Engineer,* **Universal Logic Inc.** | Charlotte, NC, USA **May 2018 – Jan 2022**

- Developed and deployed various custom Machine learning and Deep learning algorithms and reactive robot control systems for object detection, motion planning, and industrial automation tasks.
- Collected, processed, and analyzed data from diverse online and in-house sources using clustering methods, dimensionality reduction, and Python-based pipelines for robust model training.
- Developed scalable data engineering solutions using Python, C++, and AI/ML libraries such as TensorFlow, PyTorch, and Keras on AWS infrastructure, including S3 for data management and EC2 GPU instances for high-performance inference.
- Designed and maintained centralized databases and web-based interfaces for efficient data storage, access, and visualization.
- Determined optimal clustering methods and intelligence training strategies to improve robot decision-making and task execution.
- Conducted continual research on emerging ML techniques and robotics trends to refine model accuracy and system integration.

*Research Assistant,* **Real-Time & Embedded Systems Lab (mLab)** | Philadelphia, PA, USA **Jun 2017 – May 2018**

- Developed an end-to-end toolchain to streamline the process from mission planning to deployment for drone operations.
- Designed and implemented control algorithms to generate waypoints for user-defined trajectories, including multi-drone coordination, dynamic obstacle avoidance, and other advanced flight maneuvers.
- Built a simulation environment using CAD modeling integrated with ROS/Gazebo for testing and validating flight scenarios.

*CAD Application Developer,* **Quest-Global Inc.** | Bangalore, India and Bristol, United Kingdom **Jun 2011 – Oct 2013**

- Collaborated with a 25-member team to deploy a multidisciplinary cost-optimization framework for turbine blade components used in Rolls-Royce aero-engines.
- Gathered and analyzed requirements from designers, engineers, and manufacturers to develop 40+ parametric features and an intuitive Graphical User Interface (GUI) using NX design automation tools.
- Represented the Quality and Improvements team in the UK for five months, leading project timelines and co-developing geometry workflows using Simulia Isight, achieving a 100% Customer Satisfaction (C-Sat) score.
- Spearheaded two new automation projects ahead of schedule, authored 10+ technical articles for the company's Knowledge Portal (viewed over 500 times), and trained four new employees, earning Quest's prestigious Performance Recognition Award.

## EDUCATION

*Master of Science, Mechanical Engineering,* **University of Pennsylvania** **Aug 2014 - May 2016**

*Bachelor of Technology, Mechanical Engineering,* **Vellore Institute of Technology University** **Jul 2007 - Jun 2011**

## SKILLS

**Expertise:** Industrial Automation | Control Systems | Robotics | Machine Learning | Deep Learning | Project Management
**Programming Languages:** Python | R | C, C++ | C# | IEC 61131-3 (Ladder Logic, Structured Text) | LabVIEW | MATLAB
**Control Systems:** Beckhoff | TwinCAT | Siemens | TIA Portal | Allen Bradley | RSLogix 5000 | Mitsubishi PLC | MELSEC GXWorks